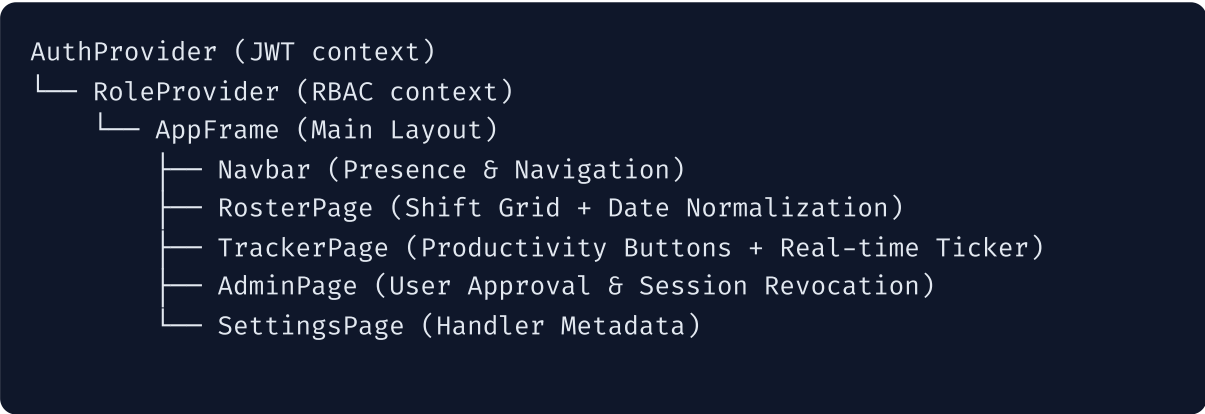


Queue Tracker — Low Level Design (LLD)

1. Frontend Component Architecture



2. API Surface

Authentication (REST)

Endpoint	Method	Description
/api/auth/register	POST	User registration with validation
/api/auth/login	POST	JWT authentication + 2FA check
/api/twoFactor/verify	POST	TOTP verification
/api/roles/users	GET	Admin only - List all users
/api/roles/delete/:id	DELETE	Admin only - Delete user

Real-Time (Socket.IO)

Event	Direction	Purpose
join_room	C → S	Connect to live updates
update_stats	C → S	Increment/Decrement INC/TASK/CALL
stats_sync	S → C	Broadcast updated counts to all users
presence_update	S → C	Broadcast online/offline status

3. Data Models (Mongoose)

User Schema

```
{
  username: { type: String, unique: true },
  password: { type: String }, // Bcrypt hashed
  role: { type: String, enum: ['Admin', 'Lead', 'Agent'] },
  isApproved: { type: Boolean, default: false },
  twoFactorSecret: { type: String }, // AES-256-GCM encrypted
  lastActive: { type: Date }
}
```

Stats Schema

```
{
  userId: { type: ObjectId, ref: 'User' },
  date: { type: String }, // ISO Date YYYY-MM-DD (IST)
  incidents: { type: Number, default: 0 },
  sctasks: { type: Number, default: 0 },
  calls: { type: Number, default: 0 }
}
```

4. Core Logic: IST Date Normalization

To ensure global consistency, all date-based logic uses the `IST_DATE` normalization:

```
const getISTDate = () => {
  return new Date().toLocaleString('en-US', {
    timeZone: 'Asia/Kolkata'
  }).split(',')[0]; // MM/DD/YYYY formatted for DB keys
};
```

5. Security Middleware

- `requireAuth` : Validates HTTP-only JWT cookie.
- `requireRole(['Admin'])` : Restricts access to sensitive routes.
- `rateLimit` : Protects login and 2FA endpoints from brute-force.